Monetary policy topics

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This session

- So far: HANK with Jacobians by hand, introduction to the SSJ toolkit
- **Now:** more advanced topics on monetary policy, using the toolkit!

    1. Cyclical income risk
    2. Maturity structure
    3. Nominal assets
    4. The stock market

Cyclical income risk

Cyclical income risk

- Recall canonical model: household takes $n_{it} = N_t$ as given and solves

$$\max_{c_{it}} \quad \mathbb{E}_0 \sum_{t=0}^{\infty} \beta_{it} \left(u(c_{it}) - v\left(n_{it}\right) \right)$$

$$c_{it} + a_{it} \leq (1 + r_t^p) a_{it-1} + (1 - \tau_t) \frac{W_t}{P_t} n_{it} e_{it}$$

$$a_{it} \geq 0$$

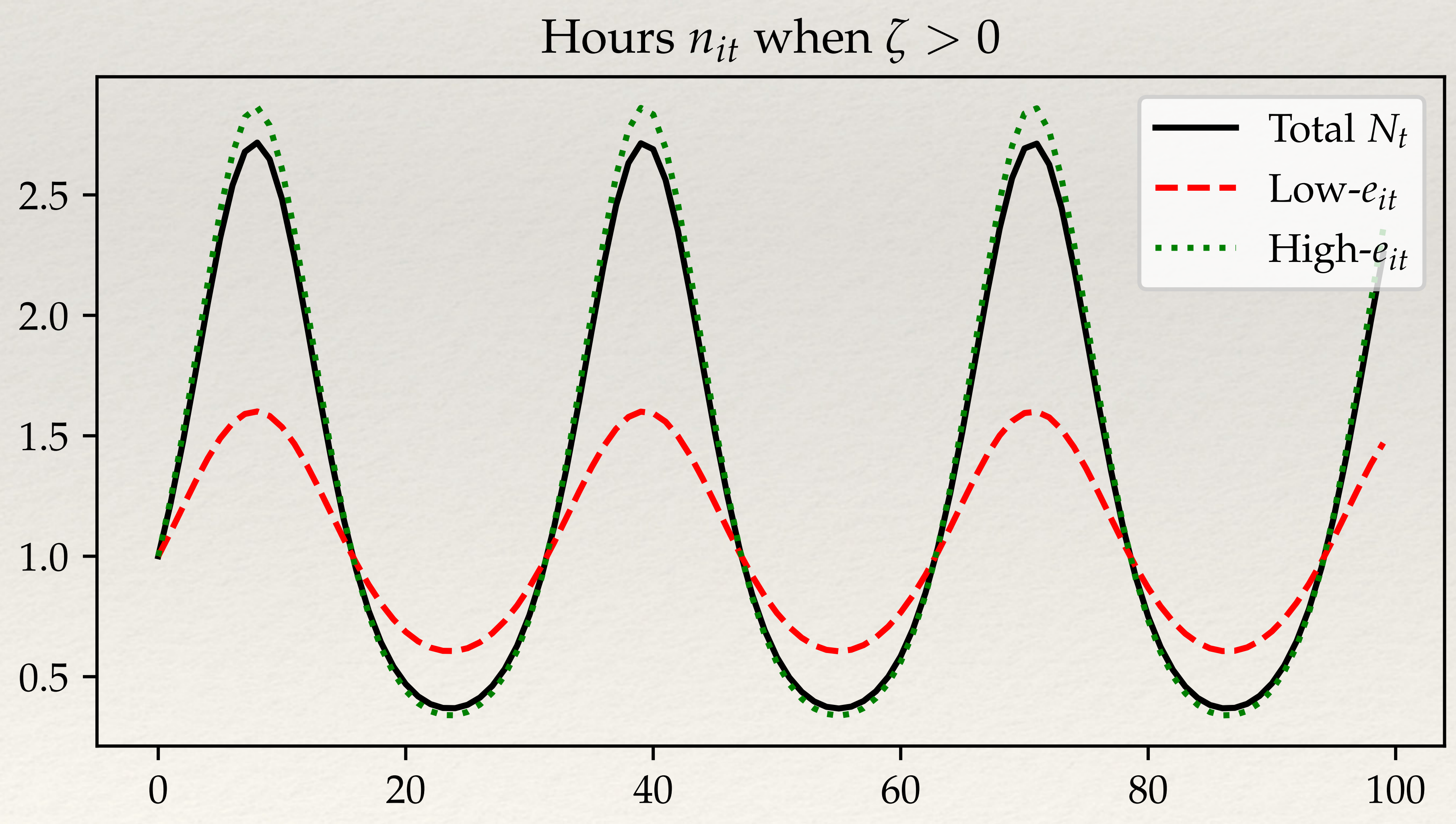
- This is restrictive: hours n_{it} don't move uniformly across people over the cycle!
- Relaxing this assumption will make a difference, but also uncover a big "puzzle"

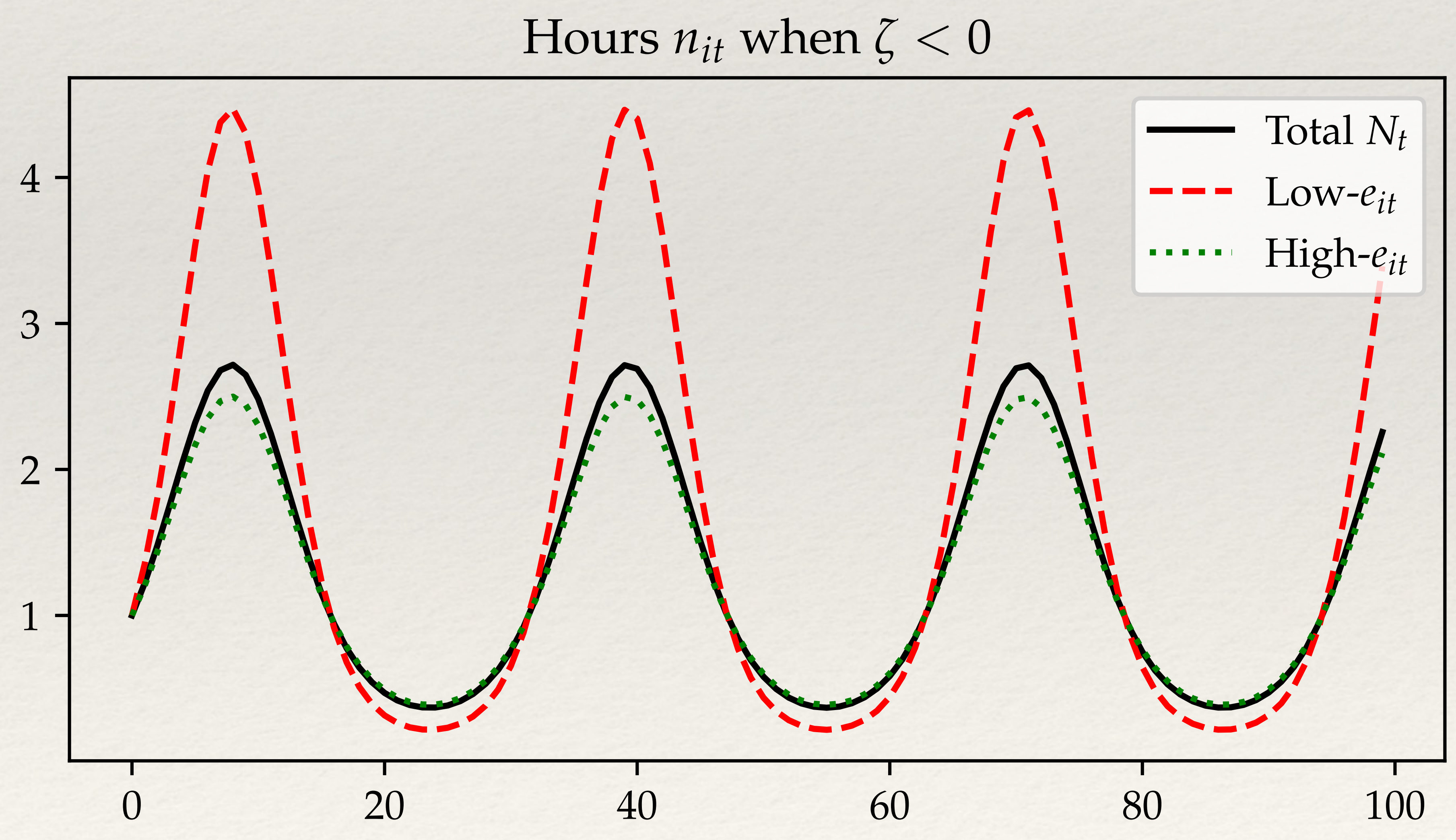
Introducing unequal rationing

- Simple way to relax is to assume that labor is rationed as a function of e_{it}, per

(assuming normalization $N_{ss} = 1$)

$$n_{it} = N_t \frac{\left(e_{it}\right)^{\zeta \log N_t}}{\mathbb{E}\left[e_i^{1+\zeta \log N_t}\right]} \equiv N_t \Gamma\left(e_{it}, N_t\right)$$





- $\zeta > 0$: procyclical inequality+income risk, $\zeta < 0$ countercyclical, $\zeta = 0$ acyclical

Introducing unequal rationing

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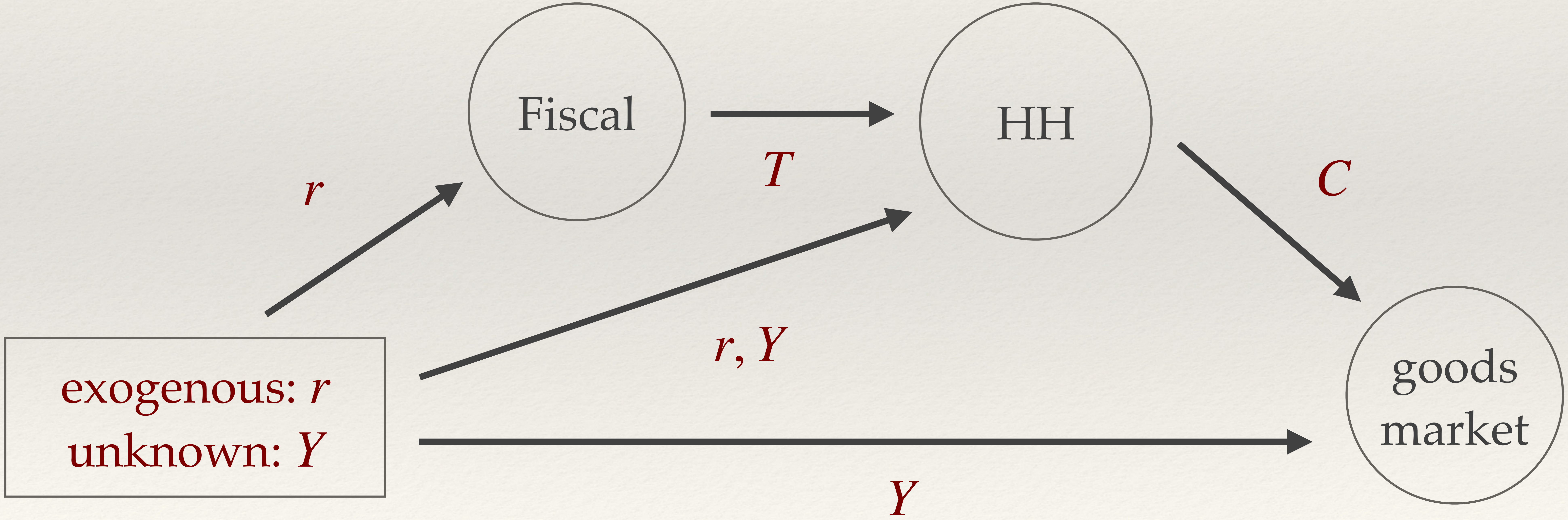
- $\zeta > 0$: procyclical inequality+income risk, $\zeta < 0$ countercyclical, $\zeta = 0$ acyclical
- In particular the distribution of pretax income $y_{it} = \frac{W_t}{P_t} n_{it} e_{it}$ moves with the cycle

$$\text{sd}\left(\log y_{it}\right) = \left(1 + \zeta \log N_t\right) \text{sd}\left(\log e_i\right)$$

- Matters because: 1) current shocks redistribute between different MPCs, and 2) future shocks change perceived income risk

How does this change the model?

- Now post-tax income z_{it} for individual i is $z_{it} = (Y_t - T_t)\Gamma\left(e_{it}, Y_t\right)$
- This results in a consumption function $\mathscr{C}_t\left(\left\{r_s, Y_s, T_s\right\}\right)$. DAG:



Practical implementation with hetinput:

- Simple modification of the “hetinput” to the household block:

```python
def income_cyclical(Y,T, e_grid, e_pdf, zeta):
    y = (Y-T) * e_grid ** (1 + zeta * np.log(Y)) / np.vdot(e_grid ** (1 + zeta * np.log(Y)), e_pdf)
    return y

household_cyc = hh.add_hetinputs([income_cyclical])

print('Household block inputs:' , household_cyc.inputs)
print('Household block outputs:' , household_cyc.outputs)
```

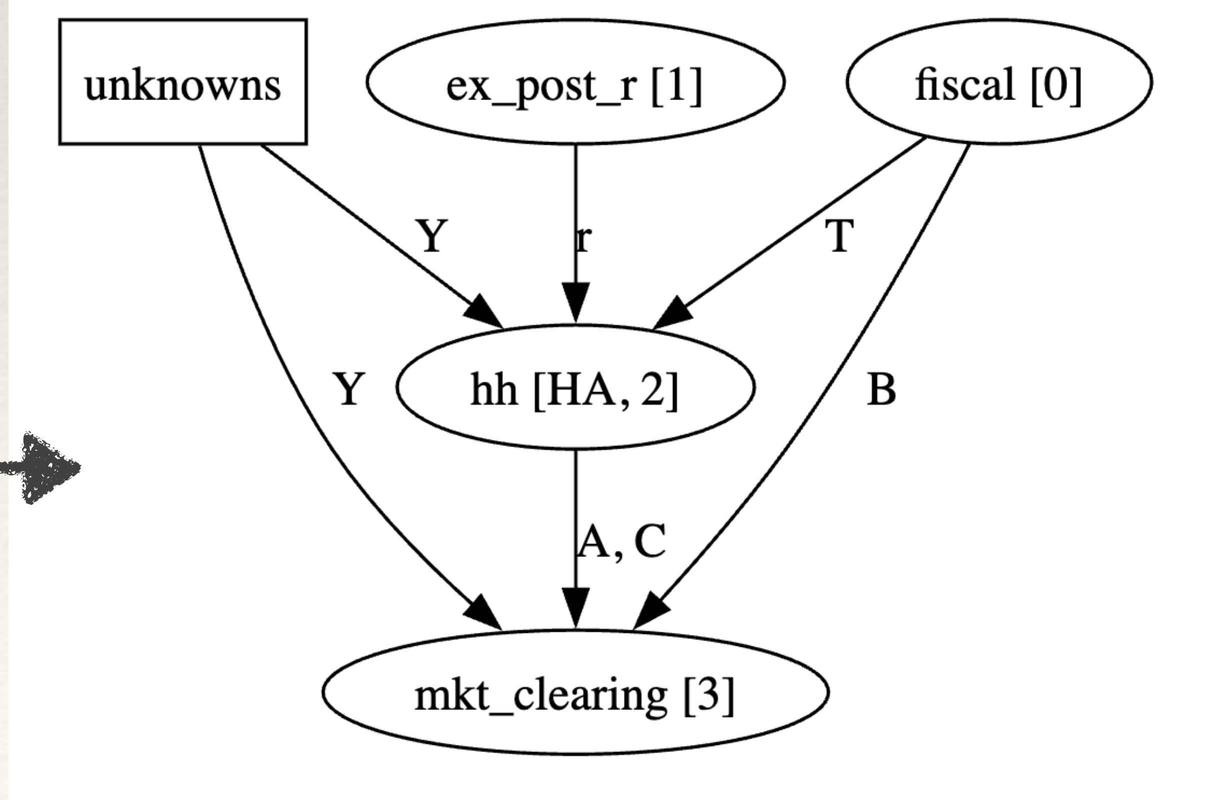
✓ 0.0s Python

```
Household block inputs: ['a_grid', 'r', 'beta', 'eis', 'Pi', 'Y', 'T', 'e_grid', 'e_pdf', 'zeta']
Household block outputs: ['A', 'C']
```

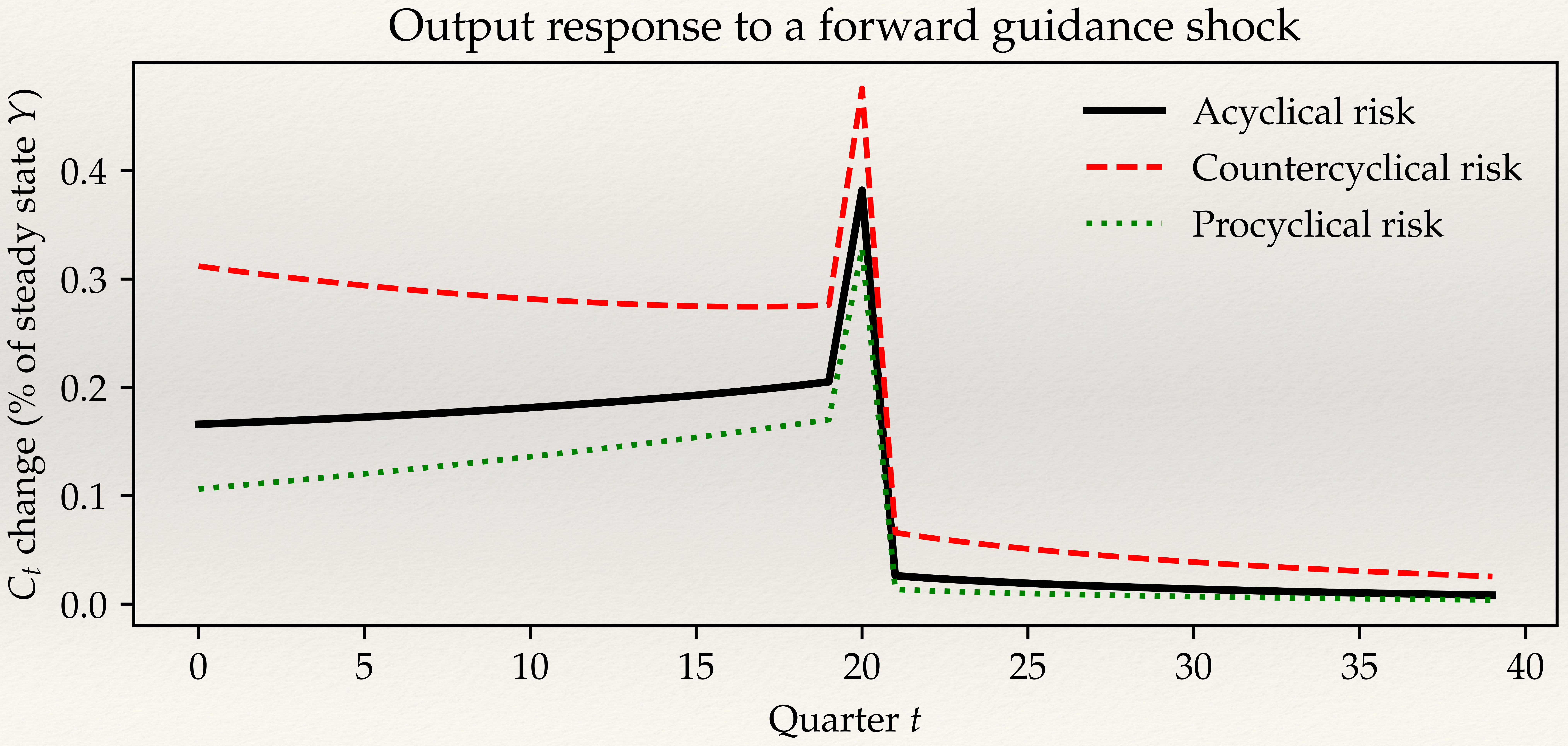
- Can then combine this and other blocks into model:

```python
model_cyc = sj.create_model([household_cyc, ex_post_r, fiscal, mkt_clearing], \
                            name="HA Model with cyclical income risk")

sj.drawdag(model_cyc, unknowns=['Y'])
```



Result: countercylical risk amplifies FG puzzle!



What is happening?

- Go back to zero liquidity limit $B_{ss} \to 0$. Now, can show:

$$C_t = \delta \cdot C_{t+1} - \frac{1}{\sigma} \cdot \text{const} \cdot \left(r_t - \bar{r}\right)$$

- There is dynamic discounting $\delta < 1$ iff procyclical income risk $\zeta > 0$
- Dynamic amplification $\delta > 1$ iff countercyclical income risk $\zeta < 0$
- Countercyclical income risk usually thought to be more plausible:
    - Theoretically: eg, from unemployment risk (Ravn-Sterk, Challe, Kekre…)
    - Empirically: eg, work by Guvenen&co and Storesletten&co
- Bilbiie's "catch-22": countercylical risk plausible at micro level, not macro!

Indirect ways to make income risk cyclical

- In richer models, income of agents typically involves multiple components,

$$y_{it} = \frac{W_t}{P_t} n_{it} e_{it} - \underbrace{T_{it}}_{\text{Taxes}} + \underbrace{Tr_{it}}_{\text{Transfers}}$$

- These also matter for cyclicality of income risk
- For example, suppose we set taxes to keep $(1 + r_t)B_t$ constant as in our baseline, but that transfers are (nontraded) dividends from firms with sticky prices
  - Both T_{it} and Tr_{it} fall when $r_t \downarrow$ (why?)
- Suppose that T_{it} are paid by the high e_{it}, but everyone owns equal share of firms
- Then income risk is procyclical! This is McKay, Nakamura, Steinsson

Maturity structure

Maturity structure

- So far: agents trade short term assets. What if longer maturities / duration?
- For tractability, assume "Calvo bonds"
  - buy one bond today for q_t, coupon payments $1, \delta, \delta^2, \ldots$
- New household problem:

$$\max_{c_{it}} \quad \mathbb{E}_0 \sum_{t=0}^{\infty} \beta_{it} \left(u(c_{it}) - v\left(N_t\right) \right)$$

$$c_{it} + q_t \lambda_{it} \leq (1 + \delta q_t) \lambda_{it-1} + Z_t e_{it}$$

$$\lambda_{it} \geq 0$$

Total number of coupons owned coming into t

- Asset pricing condition:

$$1 + r_t = \frac{1 + \delta q_{t+1}}{q_t}$$

Rewriting the problem

- Redefine $a_{it} = q_t \lambda_{it}$. Using asset pricing condition, we have

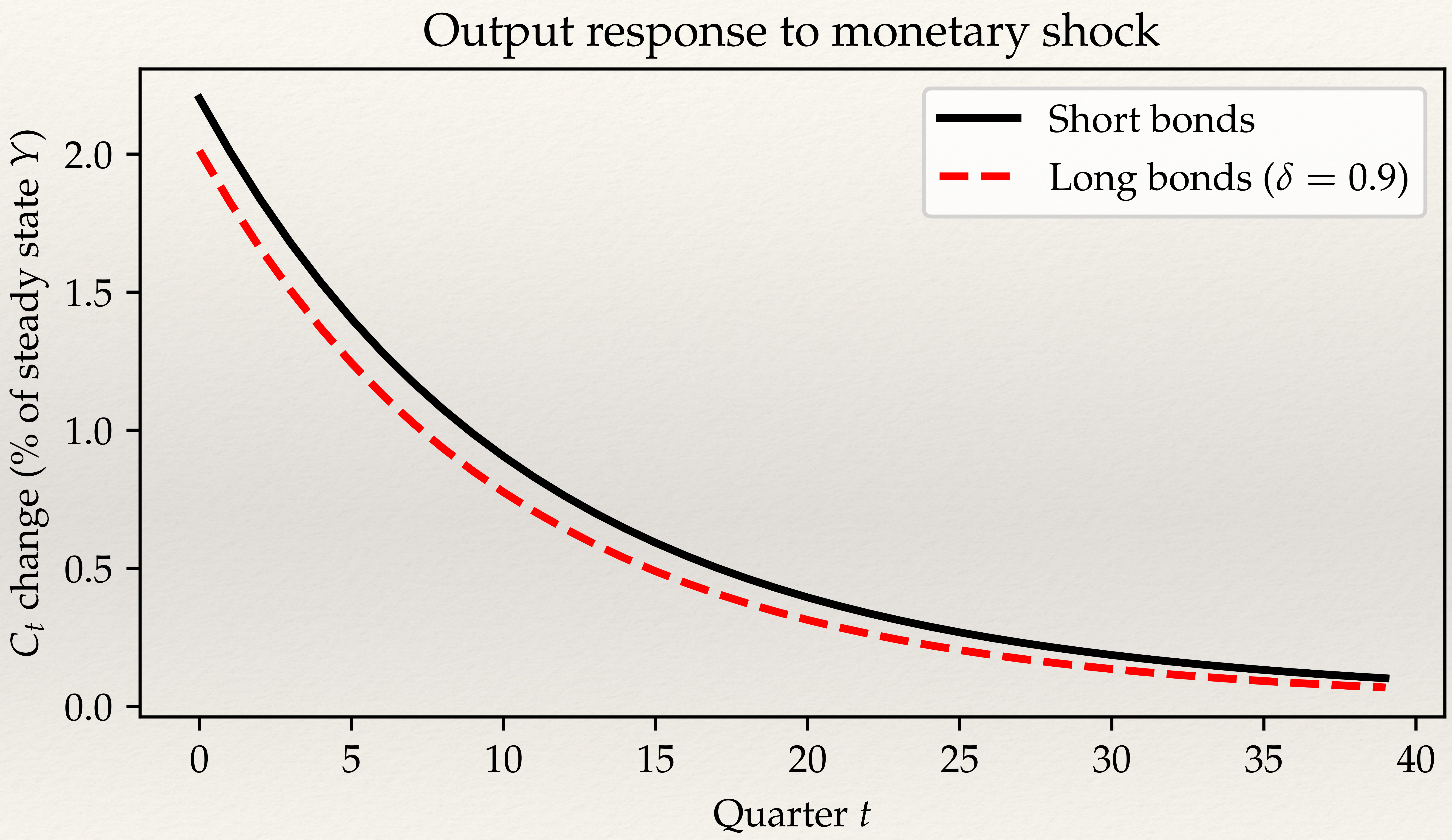
$$\max_{c_{it}} \quad \mathbb{E}_0 \sum_{t=0}^{\infty} \beta_{it} \left(u(c_{it}) - v\left(N_t\right) \right) \qquad r^p_{t+1} = r_t, t \geq 0$$

$$c_{it} + a_{it} \leq (1 + r^p_t) a_{it-1} + Z_t e_{it} \qquad r^p_0 = \frac{1 + \delta q_0}{q_{ss}}$$

$$a_{it} \geq 0$$

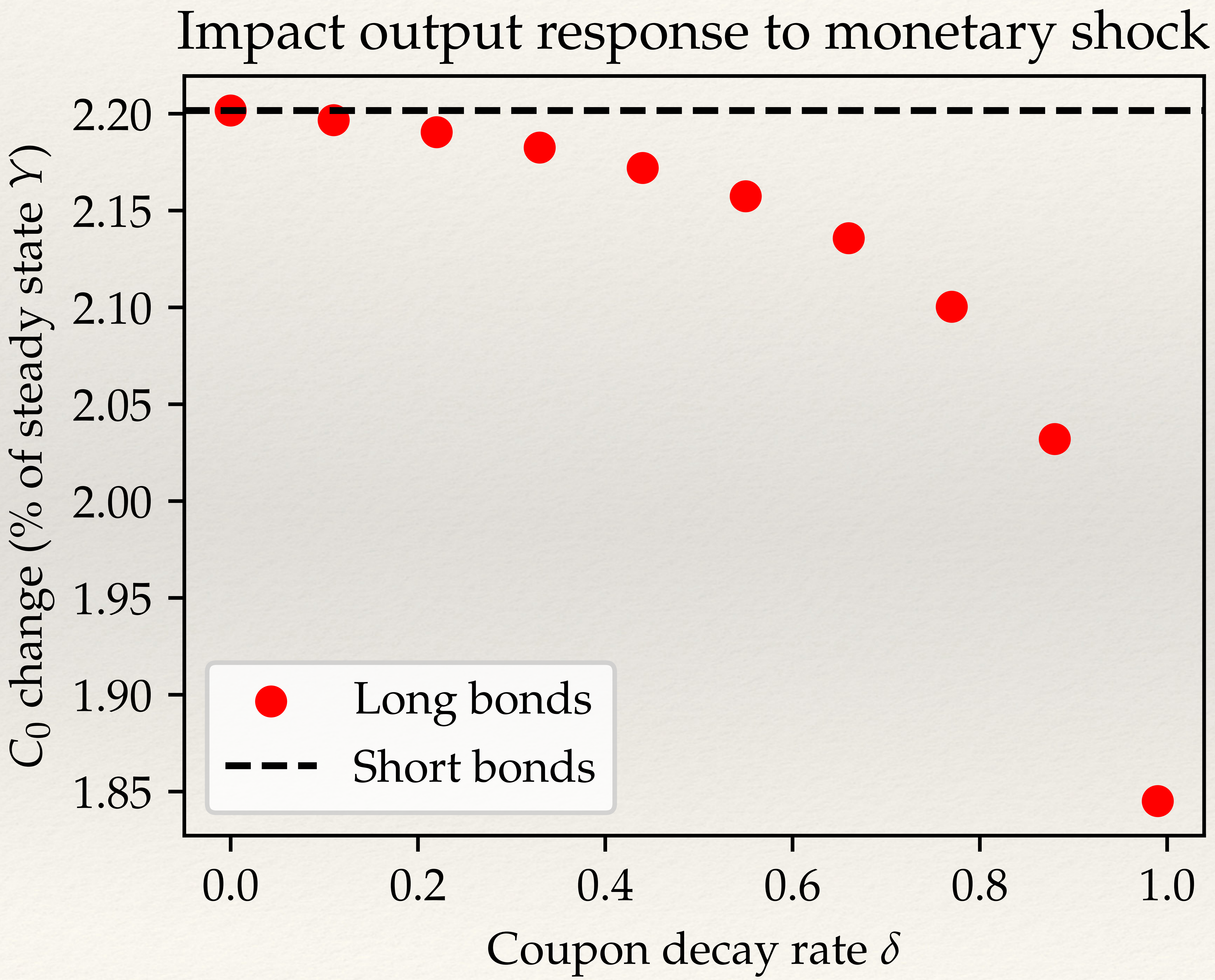
- Lower ex-ante r_t → higher bond price q_0 → higher r^p_0 (capital gain)
- Same for government budget constraint: $B_t = (1 + r^p_t) B_{t-1} + G_t - T_t$
  - here $B_t \equiv q_t \lambda_t$ is market value of gov debt. Assume **fiscal rule** $\lambda_t = \lambda_{ss}$, $G_t = G_{ss}$

Effect of long maturities on monetary impulse



- Longer maturities reduce the output effect of a monetary shock ! Why?

Effect of maturity structure explained



- $\delta \uparrow$: households make cap. gains from monetary expansion, gov makes loss
- Redistribution from taxpayers to bond holders. Latter have lower MPCs, so aggregate effect on spending falls!
- Expect monetary policy to be more powerful in countries with short durations (eg ARMs)
    - Actually true in data! (Calza et al)

Nominal assets

Nominal assets

- So far, all assets are real. But in practice, many assets are nominal
  - Mortgages, bonds, etc
  - Creates very large exposures to inflation risk via nominal positions (see e.g. Doepke and Schneider)
- Consider 1-period bonds:

$$\max_{c_{it}} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta_{it} \Big(u(c_{it}) - v\left(N_t\right) \Big)$$

$$P_t c_{it} + A_{it} \leq (1 + i_t) A_{it-1} + P_t Z_t e_{it}$$

$$A_{it} \geq 0$$

Can relax to have $P_t \underline{a}$

- Fisher equation:

$$1 + r_t = \frac{(1 + i_t) P_t}{P_{t+1}}$$

In practice, borrowing constraints interact with inflation in complex ways! ("Tilt effect")

Rewriting the problem

- Now redefine A_{it}/P_t. Using Fisher equation, we have

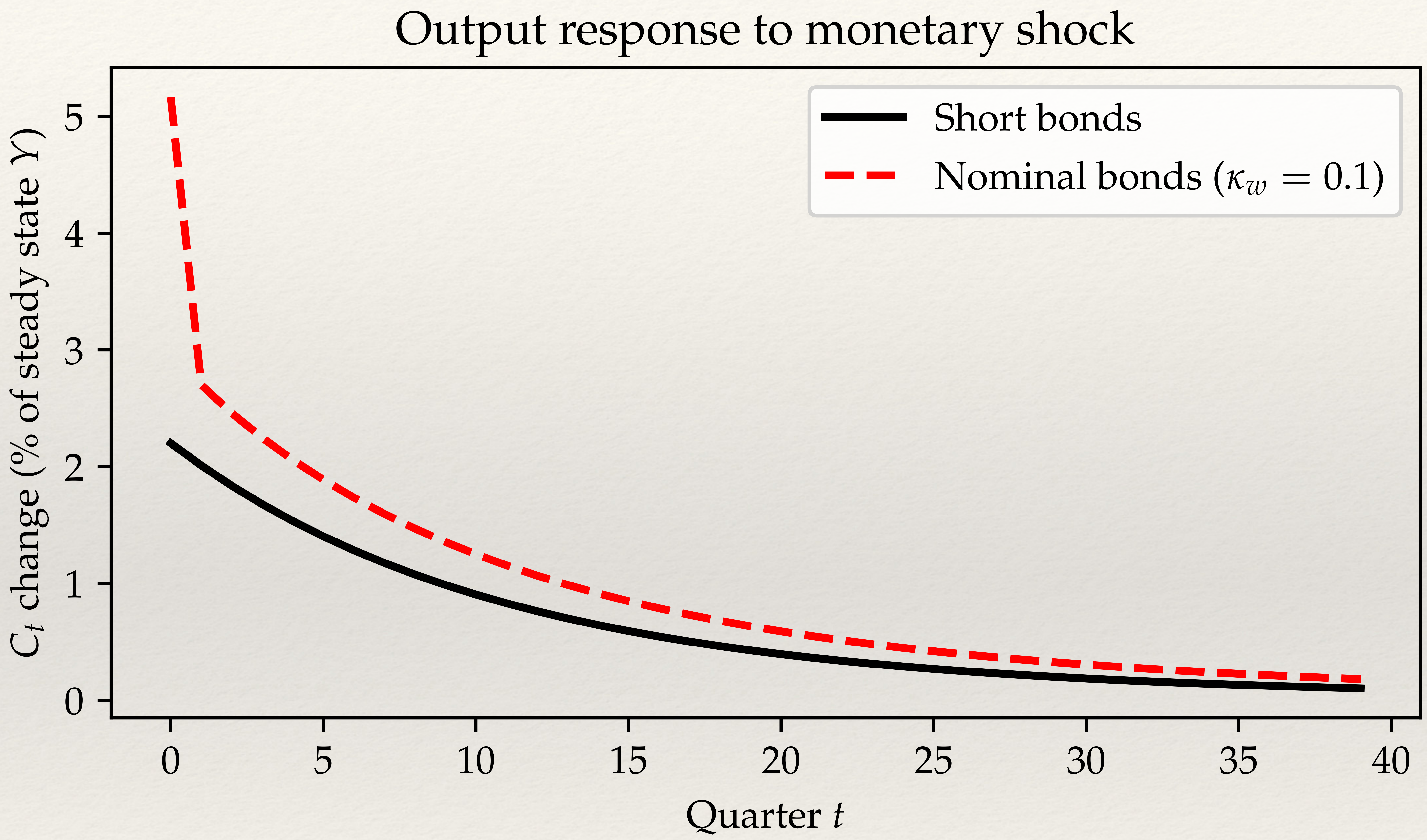
$$\max_{c_{it}} \quad \mathbb{E}_0 \sum_{t=0}^{\infty} \beta_{it} \left(u(c_{it}) - v\left(N_t\right) \right) \qquad r_{t+1}^p = r_t, t \geq 0$$

$$c_{it} + a_{it} \leq (1 + r_t^p) a_{it-1} + Z_t e_{it} \qquad r_0^p = \left(1 + r_{ss}\right) \frac{P_{ss}}{P_0} = \left(1 + r_{ss}\right) \frac{1 + \pi_{ss}}{1 + \pi_0}$$

$$a_{it} \geq 0$$

- Same for gov budget constraint, fiscal rule continues to keep $(1 + r_t)B_t$ constant
- Even with r_t rule, inflation now matters for aggregate demand due to nominal revaluation! Use Phillips curve: $\pi_t = \kappa_w(Y_t - 1) + \beta\pi_{t+1}$

Output effect of a monetary policy shock



- **Fisher effect**: inflation redistributes from bondholders to taxpayers
- Those agents have higher MPCs, this boosts demand (effect bigger with steeper P.C.)

The stock market

Introducing a stock market

- Calibration with $A_{ss} = B_{ss} = Y$ not realistic
  - Households hold a lot more wealth than this, ~ 5 GDP in the U.S.
  - Invested (at least) in stocks **in addition** to bonds
- To incorporate this, change calibration to have markups $\mu > 0$, $A_{ss}/Y = 5$
- Firm problem now implies $\dfrac{W_t}{P_t} = \dfrac{1}{\mu}$ and dividends $d_t = (1 - \tau_t)\left(1 - \dfrac{1}{\mu}\right) Y_t$
- Stocks are claim to dividends with price p_t
- A mutual fund owns bonds and stocks, sells shares to hh's with return r_t^p

[Households owning stocks and bonds directly similar, see portfolio lecture]

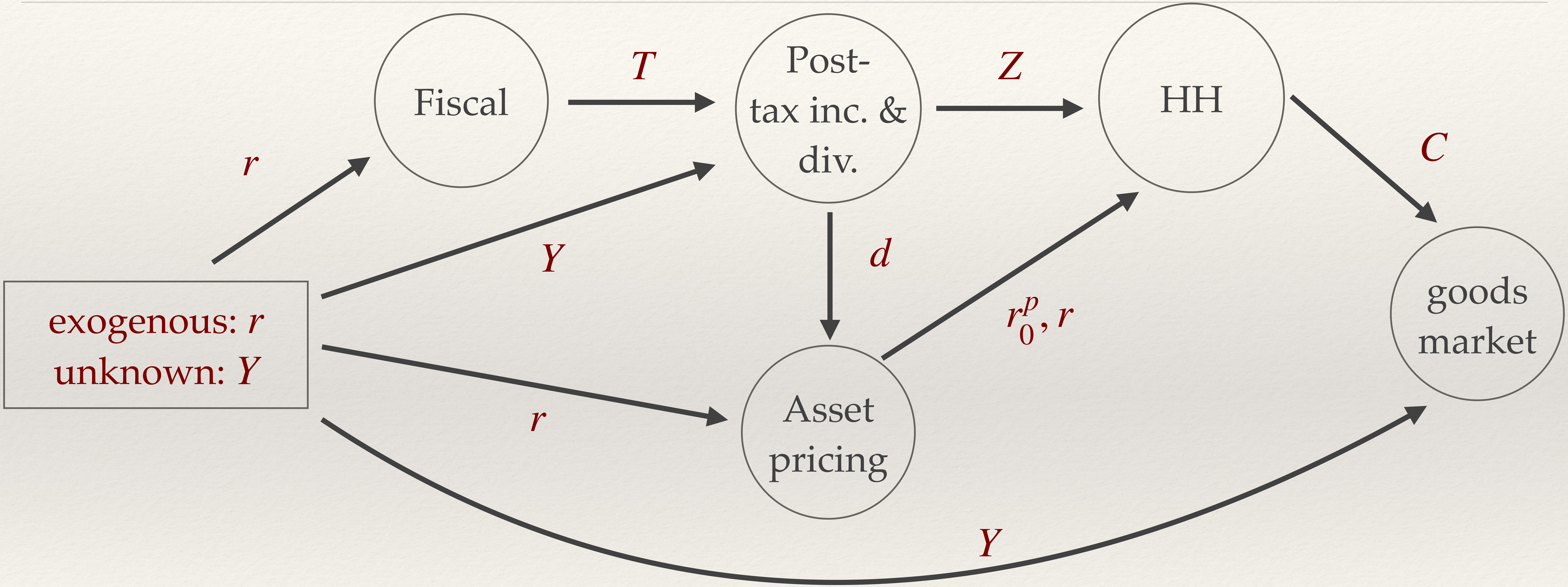
Asset pricing conditions

- No arbitrage condition for the mutual fund: $1 + r_t = \dfrac{p_{t+1} + d_{t+1}}{p_t}$, for $t \geq 0$
- Ex-post return to mutual fund holders between t and $t+1$: $r^p_{t+1} = r_t$
- What is the ex-post return between $t = -1$ and $t = 0$?
  - Stock market price at date 0 is $p_0 = \displaystyle\sum_{s=1}^{\infty}\left(\prod_{u=0}^{s}\frac{1}{1+r_u}\right)d_s$

    in ss, just $p_{ss} = \dfrac{d_{ss}}{r}$
  - Total ex-post return is $1 + r^p_0 = (1 + r_{ss})\omega + \dfrac{p_0 + d_0}{p_{ss}}(1 - \omega)$

    s.s. share of bonds in mutual funds portfolio B_{ss}/A_{ss}

DAG of model with traded markups

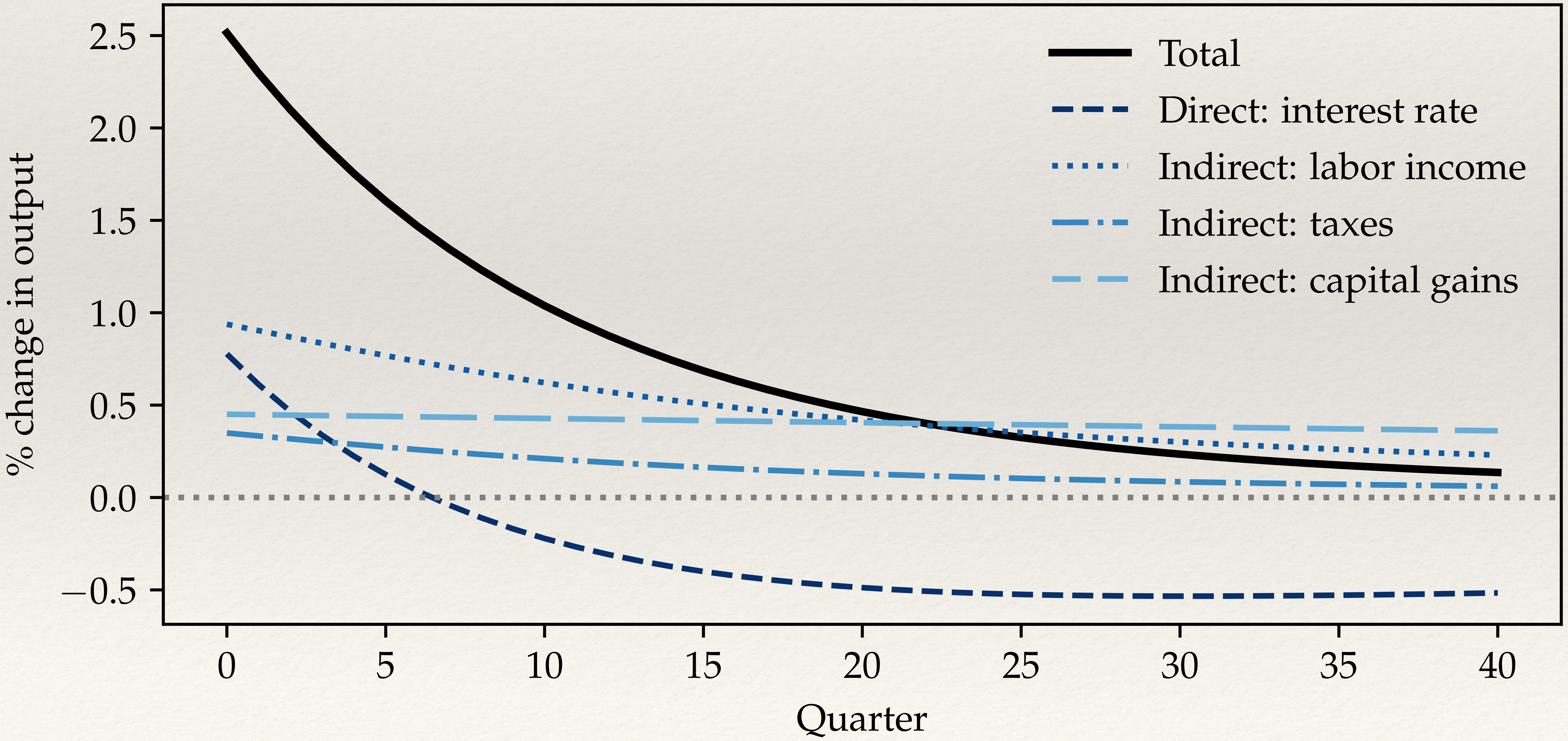


- Goods market clearing: $Y_t = \mathscr{C}_t\left(r_0^p, \{r_s\}, \{(Y_s - T_s)/\mu\}\right) + G_{ss}$

Impulse response to AR(1) monetary shock

- Calibration: $B_{ss}/Y_{ss} = 1$

$$d\mathbf{Y} = \underbrace{\mathbf{M}^r d\mathbf{r}}_{\text{Direct interest rate effect}} - \underbrace{\mathbf{M}/\mu B_{ss} d\mathbf{r}}_{\text{Tax effect}} + \underbrace{\mathbf{M}/\mu d\mathbf{Y}}_{\text{Labor income effect}} + \underbrace{\mathbf{m} A_{ss} dr_0^p}_{\text{Cap. gain effect}}$$



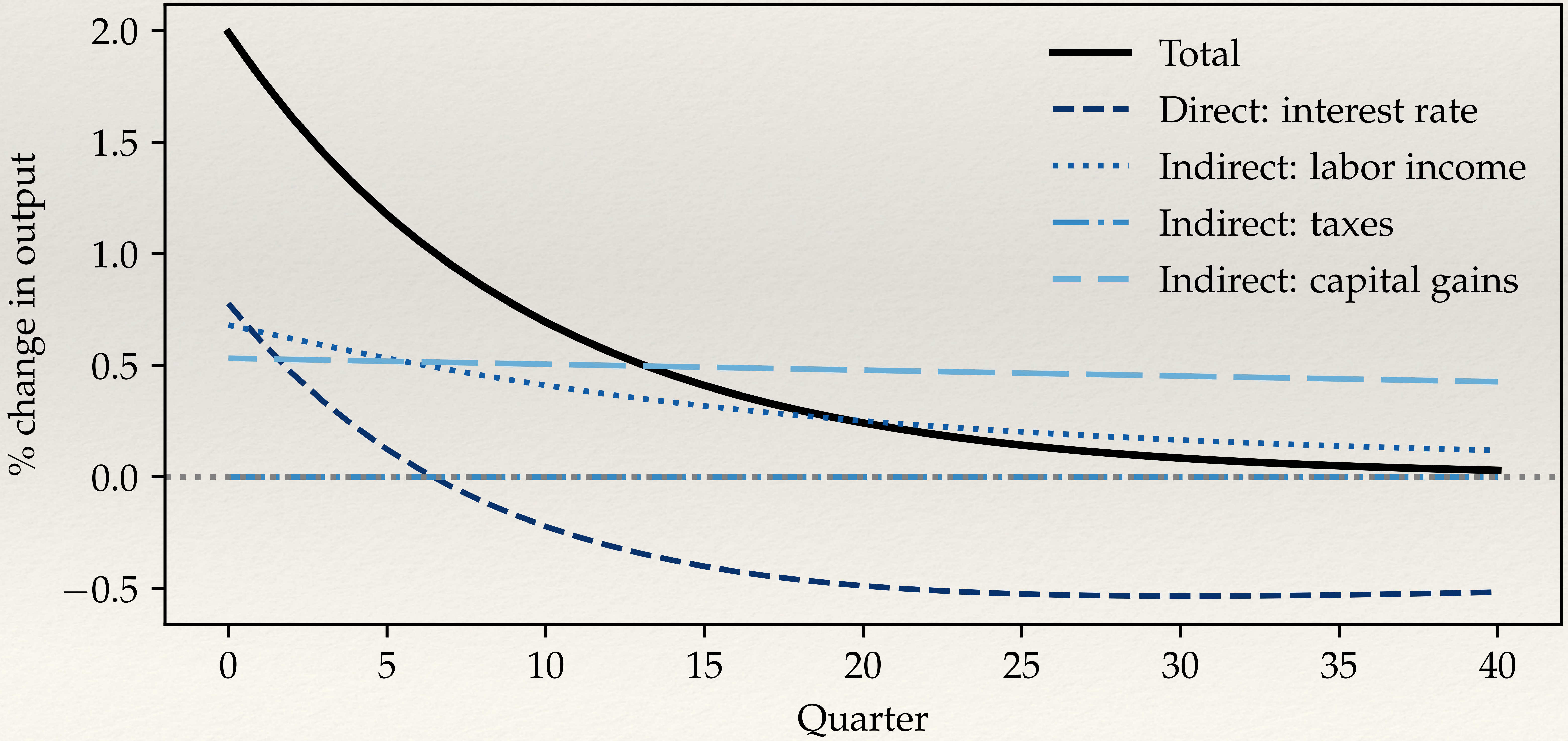
New capital gain effect!

Magnitude governed by **m**, the MPC out of capital gains (low in data & model, here about 1 cent on the dollar per quarter)

Impulse response in model without bonds

- Calibration: $B_{ss}/Y_{ss} = 0$

$$d\mathbf{Y} = \underbrace{\mathbf{M}^r d\mathbf{r}}_{\text{Direct interest rate effect}} + \underbrace{\mathbf{M}/\mu d\mathbf{Y}}_{\text{Labor income effect}} + \underbrace{\mathbf{m}A_{ss}dr_0^p}_{\text{Cap. gain effect}}$$

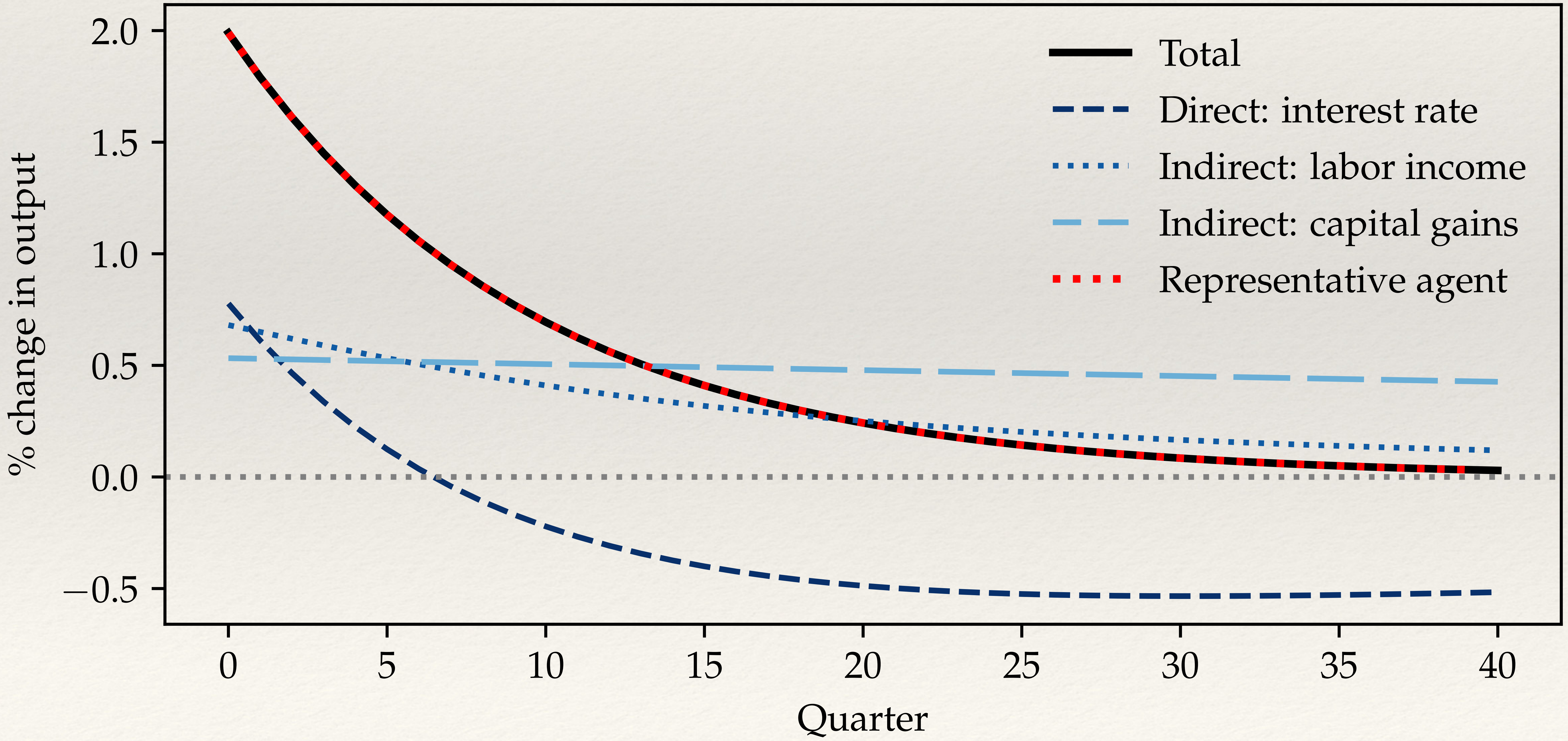


Tax effect is now zero…

Impulse response in model without bonds

- Calibration: $B_{ss}/Y_{ss} = 0$

$$d\mathbf{Y} = \underbrace{\mathbf{M}^r d\mathbf{r}}_{\text{Direct interest rate effect}} + \underbrace{\mathbf{M}/\mu d\mathbf{Y}}_{\text{Labor income effect}} + \underbrace{\mathbf{m} A_{ss} dr_0^p}_{\text{Cap. gain effect}}$$



Total effect is exactly the same as in RANK!

$$dC_t = 0.25 \cdot C \cdot \frac{\rho^t}{1-\rho}$$

Coincidence?

Neutrality result

- No! For the calibration with $B_{ss} = 0$ and $u(c) = \log c$, we prove (see ARE):

$$\color{blue}{d\mathbf{Y}} = -\,C\mathbf{U}d\mathbf{r} \qquad \mathbf{U} \equiv \begin{pmatrix} 1 & 1 & 1 & 1 & \cdots \\ 0 & 1 & 1 & 1 & \cdots \\ 0 & 0 & 1 & 1 & \cdots \\ 0 & 0 & 0 & 1 & \cdots \\ \vdots & \vdots & \vdots & \vdots & \ddots \end{pmatrix}$$

- Monetary policy effects in HANK and RANK are the same for **any** $d\mathbf{r}$ **!**
- **Werning (2015) result version 2** (log utility + "acyclical liquidity")

Summary

- HANK substantially enriches the analysis of monetary policy
- Key points:
    - Countercyclical income risk has large amplification effects
    - Maturity structure important due to capital gains-induced redistribution
    - Nominal positions relevant due to inflation-induced redistribution
    - With traded stocks, can get other instance of equivalence to rep agent
- Topics we left out (see next lecture+ARE paper): monetary-fiscal interactions, fiscal theory of the price level, investment, Taylor rules…